

Appendix C: replay transcripts and expected markers

Repository source: `proof/APPENDIX_C_REPLAY_TRANSCRIPTS.md`

Status: summary of archived evidence **Transcript authority:** checksummed files under `certificate/transcripts/`
This appendix does not replace the transcript files. It identifies what each file records, what a successful run must contain, and how the records should be interpreted.

C.1 Transcript inventory

File	Path exercised	Required terminal marker
<code>exact_certificate.txt</code>	Production exact-rational verifier	MINVOL UNIVERSAL CONTACT-BOUND CERTIFICATE: EXACT
<code>binary64_audit.txt</code>	Independent NumPy binary64 audit	MINVOL UNIVERSAL CONTACT-BOUND BINARY64 AUDIT: PASS
<code>python_unittest.txt</code> <code>adversarial_mutations.txt</code>	Ten reconstruction/regression tests Seven deliberate corruptions	Ran 10 tests followed by OK MINVOL ADVERSARIAL MUTATION SUITE : PASS
<code>base_r_audit.txt</code>	Independent base-R audit	MINVOL UNIVERSAL CONTACT-BOUND BASE-R AUDIT: PASS
<code>export_equivalence.txt</code>	Private-source/public-export theorem-field comparison performed during construction	PUBLIC EXPORT EQUIVALENCE: PASS
<code>environment.txt</code>	Construction-environment versions and platform	no theorem marker; provenance only

Exact SHA-256 values for these files are recorded in `certificate/SHA256SUMS.txt` and the root `SHA256SUMS.txt`.

C.2 Exact transcript

The exact transcript should report, among other values:

```
split radius:          0.61139120200000000000
weakest lower row:      03B02
changed fan vertices:    18229
minimum exact fan margin: 0.0000000002888904207092857876733697822441644655792123744
near-Jung handoff margin: 0.0000021074345301574846190550703110031602996750911392740
universal coefficient lower: 0.4118775637803022742481789259828398710025748462041567278
```

The exact output is a human-readable projection of integer and rational computations. The production program also requires byte-identical reconstruction of the authoritative certificate and exact audit artifacts.

C.3 Binary64 transcript

The numerical audit should reconstruct the same geometry with ordinary floating-point operations and report values agreeing at the documented scale. It is expected to differ in the last displayed digits from exact decimal renderings.

A pass means the independent numerical implementation agrees within its stated tolerances. It is not a substitute for the exact finite proof.

C.4 Base-R transcript

The base-R implementation was run on the named author's Mac. It reports the four refined rows, fan census and minimum margin, determinant lower bound, coordinate widths, cap totals, disjointness gap, near-Jung value, theorem coefficient, and predecessor gain.

The archived transcript did not record the exact R version. This omission is reported rather than filled by inference. Hosted continuous integration uses the declared R version in `.github/workflows/verification.yml`.

C.5 Mutation transcript

A successful mutation transcript shows seven tests and ends with:

```
Ran 7 tests
OK
MINVOL ADVERSARIAL MUTATION SUITE: PASS
```

The test succeeds only when every corrupted object is rejected. A mutation that passes through the verifier is a test failure, even if the final coefficient is numerically unchanged.

C.6 Export-equivalence transcript

The standalone public package was compared semantically with the authoritative private theorem package during construction. Ten theorem-critical fields were checked, including the coefficient, split, lower rows, bottleneck, fan minimum, core, cap total, near-Jung value, and handoff margin.

This transcript records construction provenance. The public release does not require access to the private repository to replay its own proof.

C.7 Reproducing the records

Run the Python paths:

```
./run_python_checks.sh
```

Run the independent R path:

```
./run_r_checks.sh
```

Run both language paths:

```
./run_all.sh
```

To refresh a transcript deliberately, capture the program output, review it, refresh checksum manifests, and commit the source and new record together. Never edit a transcript to manufacture a pass marker.

C.8 Interpretation

The transcript set demonstrates repeatability of the finite programs on the recorded environments. It does not demonstrate:

- external subject-matter review;
- novelty;
- correctness of every mathematical reduction outside the finite checker;
- immunity to a common bug shared by multiple implementations;
- peer-reviewed publication.

Those distinctions must remain visible in any release announcement.